

Grade 6 Integrated Mathematics Comprehensive Diagnostic Exam (Set 3)

Name: _____ Class: _____ Student ID: _____ Time: 60 Mins Total: 100 Pts Score: _____

Section 1: Concepts & Fill-in-the-Blank (4 pts each, 24 pts total)

Score: _____

1. Given the set of numbers: 1, 2, 4, 6, 9, 13, 21, 29, 36, 49:

- (a) Prime numbers: _____
 (b) Square numbers: _____
 (c) The number that is neither prime nor composite: _____

2. Let $A = 2^3 \times 3 \times 5$ and $B = 2^2 \times 3^2 \times 7$:

- (a) Find $\text{HCF}(A, B)$ in prime power form: _____;
 (b) Find $\text{LCM}(A, B)$: _____.

3. [Divisibility Property] Which of the following statements is TRUE? ()

- A. A multiple of any positive integer is always strictly greater than its factor.
 B. All even integers are composite numbers.
 C. Any integer divisible by 9 is always divisible by 3.
 D. The product of any two prime numbers is always prime.

4. Convert the improper fraction $\frac{29}{6}$ to a mixed number: _____; Convert the mixed number $3\frac{4}{7}$ to an improper fraction: _____.

5. Evaluate: $2^3 + 3^2 =$ _____; Simplify $10^3 \times 10^4 = 10$ —.

6. [Bilingual Terminology Matching] Match the English math terms with their meanings:

- | | | | |
|----------------------|---------------------------|----------------------------------|--|
| ① Prime Factor (LCM) | A. Least Common Multiple | ③ LCM ($\frac{a}{b}$) | C. Inverted fraction ($\frac{b}{a}$ for $\frac{a}{b}$) |
| ② Reciprocal number | B. Factor that is a prime | ④ Divisibility without remainder | D. Capable of being divided |

Section 2: Formal Calculation Steps (6 pts each, 24 pts total)

Score: _____

7. Write 72 as a product of its prime factors in index notation:

72 = _____

8. Evaluate: $\frac{5}{12} + \frac{7}{18}$ (Must show common denominator steps):

Solution:

9. Evaluate: $3\frac{1}{4} - 1\frac{5}{6}$ (Convert to improper fractions or borrow):

Solution:

10. Evaluate using mental calculation tricks: $3.6 \div 0.4 + 1.8 \times 5$:

Solution:

Section 3: Smart Comparisons & Applied Problem (22 pts total)

Score: _____

11. Compare the following pairs of fractions (Show your strategic reasoning): (4 pts each, 12 pts total)

- (a) $\frac{13}{20}$ and $\frac{13}{24}$ (b) $\frac{19}{21}$ and $\frac{23}{25}$ (c) $\frac{5}{6}$ and $\frac{7}{9}$

- (a) _____
(b) _____
(c) _____

12. [Real-World Application] A decorative ribbon is 10 meters long. First, $\frac{2}{5}$ of the entire ribbon is cut off. Then, an additional $1\frac{1}{2}$ meters is used. How many meters of ribbon remain? (10 pts)

Solution: _____

_____**Section 4: School Integrated Track Challenge Problems (30 pts total)**

Score: _____

13. [Divisibility & Modulo Dual-Variable Problem] (15 pts)A six-digit integer $\overline{52A74B}$ is divisible by both 4 and 9.

- (a) If the number is divisible by 4, what are the possible values for digit B ?
(b) If the number is also a multiple of 5, find the unique value of B and the corresponding value(s) of digit A .

(a) Because $\overline{4B}$ is divisible by 4, $B \in$ _____.

(b) Since it is also a multiple of 5, B must be _____.

Sum of digits $S = 5 + 2 + A + 7 + 4 + B =$ _____.

For S to be divisible by 9, $A =$ _____.

Answer: $A =$ _____, $B =$ _____, the six-digit number is _____.

14. [24-Game Factor Reorganization & Powers Exploration] (15 pts)

(a) Use the four numbers 3, 3, 8, 8 along with $+$, $-$, $\times$, $\div$ and parentheses to create an arithmetic expression equal to 24:

(b) Let 2^n denote the product of n factors of 2. What is the smallest positive integer n such that $2^n > 1000$?

(a) $24 =$ _____

(b) Because $2^8 =$ _____, $2^9 =$ _____, $2^{10} =$ _____,
therefore the smallest positive integer $n =$ _____.



[Diagnostic Test] Answer Key & Detailed Rubric

Section 1: Concepts (24 pts, 4 pts each)

- (a) Primes: 2, 13, 29; (b) Square numbers: 1, 4, 9, 36, 49; (c) 1.
- (a) $\text{HCF} = 2^2 \times 3 = 12$; (b) $\text{LCM} = 2^3 \times 3^2 \times 5 \times 7 = 2520$.
- C.
- $4\frac{5}{6}$; $\frac{25}{7}$.
- $8 + 9 = 17$; $10^{3+4} = 10^7$.
- ①-B, ②-C, ③-A, ④-D.

Section 2: Formal Steps (24 pts, 6 pts each)

- $72 = 2^3 \times 3^2$ (Method 4 pts, index form 2 pts).
- $\text{LCM}[12, 18] = 36$; Expression = $\frac{15}{36} + \frac{14}{36} = \frac{29}{36}$.
- Expression = $\frac{13}{4} - \frac{11}{6} = \frac{39}{12} - \frac{22}{12} = \frac{17}{12} = 1\frac{5}{12}$.
- Expression = $9 + 9 = 18$.

Section 3: Smart Comparisons & Application (22 pts)

- (a) $\frac{13}{20} > \frac{13}{24}$ (Same numerator, smaller denominator is greater); (b) $\frac{19}{21} < \frac{23}{25}$ (Residual to 1: $1 - \frac{19}{21} = \frac{2}{21} > 1 - \frac{23}{25} = \frac{2}{25}$); (c) $\frac{5}{6} = \frac{15}{18} > \frac{7}{9} = \frac{14}{18}$.
- First cut: $10 \times \frac{2}{5} = 4$ m; Remaining: $10 - 4 - 1.5 = 4.5$ m (or $4\frac{1}{2}$ m).

Section 4: Challenge Problems (30 pts)

- (a) $B \in \{0, 4, 8\}$ (5 pts); (b) Since it is a multiple of 5, $B = 0$ (3 pts). Sum of digits = $5 + 2 + A + 7 + 4 + 0 = 18 + A$. For $18 + A$ to be divisible by 9, A can be 0 or 9 (5 pts). Numbers are 520740 or 529740 (2 pts).
- (a) $8 \div (3 - 8 \div 3) = 8 \div \frac{1}{3} = 24$ (8 pts); (b) $2^8 = 256$, $2^9 = 512$, $2^{10} = 1024$. Smallest $n = 10$ (7 pts).